Foundation Models for Visual Brain Encoding

CSAI Assignment 2 Report

Name: Evan Bijoy
Roll Number: 2023101080

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1. Introduction

This is an analysis of visual brain encoding using deep neural networks and foundation models. We investigate how different model architectures align with neural responses in the human visual cortex, using fMRI data from the Algonauts 2023 Challenge.

The study is organized into three parts:

1. **CNN Baseline:** Comparing ResNet-50, EfficientNet-B0, and SqueezeNet
2. **VLM Encoding:** Evaluating CLIP and BLIP with visual, semantic, and joint features
3. **Layer-wise Analysis:** Testing hierarchical correspondence using DINOv2

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2. Dataset and ROI Selection

2.1 Dataset

- **Source:** Algonauts 2023 Challenge (Natural Scenes Dataset)
- **Subject:** 02
- **Training images:** 9,841
- **Test images:** 159
- **fMRI dimensions:** LH (19,004 vertices), RH (20,544 vertices)

Some images from the dataset:

2.2 ROI Selection

Assigned ROI (d) - Place-selective regions (floc-places):

ROI	Full Name	LH Vertices	RH Vertices	Function
OPA	Occipital Place Area	1,494	2,434	Local scene geometry

ROI	Full Name	LH Vertices	RH Vertices	Function
PPA	Parahippocampal Place Area	1,272	1,490	Scene categories
RSC	Retrosplenial Cortex	481	683	Spatial navigation

Choice ROI (a) - Early retinotopic regions (prf-visualrois):

ROI	Description	LH Vertices	RH Vertices	Function
V1v/V1d	Primary visual cortex	599/845	616/677	Edge detection
V2v/V2d	Secondary visual cortex	653/620	883/623	Texture processing
V3v/V3d	Tertiary visual area	553/574	732/756	Form processing
hV4	Human V4	606	656	Color, shape

Rationale: Early retinotopic areas (V1-V4) provide excellent contrast with place-selective regions, allowing us to test the hierarchical hypothesis that early DNN layers align with early visual areas while later layers align with category-selective regions.

Vertex Selection: 10 random vertices per ROI from the left hemisphere were selected for model training.

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3. Part 1: CNN-Based Encoding Models

3.1 Models Evaluated

Model	Architecture	Parameters	Feature Dim	Key Characteristics
ResNet-50	Residual Network	25.6M	2,048	Skip connections, deep
EfficientNet-B0	Compound Scaling	5.3M	1,280	Balanced depth/width
SqueezeNet	Fire Modules	1.2M	512	Lightweight, fast

3.2 Methodology

- **Feature Extraction:** Final pooling layer (before classification head)
- **Encoding Model:** Ridge regression ($\alpha = 1000$)
- **Evaluation:** Pearson correlation between predicted and actual fMRI responses

3.3 Results

Place-Selective ROIs (floc-places)

Model	OPA	PPA	RSC	Mean
ResNet-50	**0.445**	**0.555**	**0.507**	**0.502**
EfficientNet-B0	0.389	0.509	0.486	0.461
SqueezeNet	0.331	0.476	0.425	0.411

Early Retinotopic ROIs (prf-visualrois)

Model	V1v	V1d	V2v	V2d	V3v	V3d	hV4	Mean
ResNet-50	0.286	0.318	0.307	0.292	0.239	0.296	**0.459**	0.314
EfficientNet-B0	0.236	0.251	0.213	0.208	0.140	0.223	0.394	0.238
SqueezeNet	**0.414**	**0.444**	**0.360**	**0.395**	**0.302**	**0.440**	0.517	**0.410**

3.4 Visualization

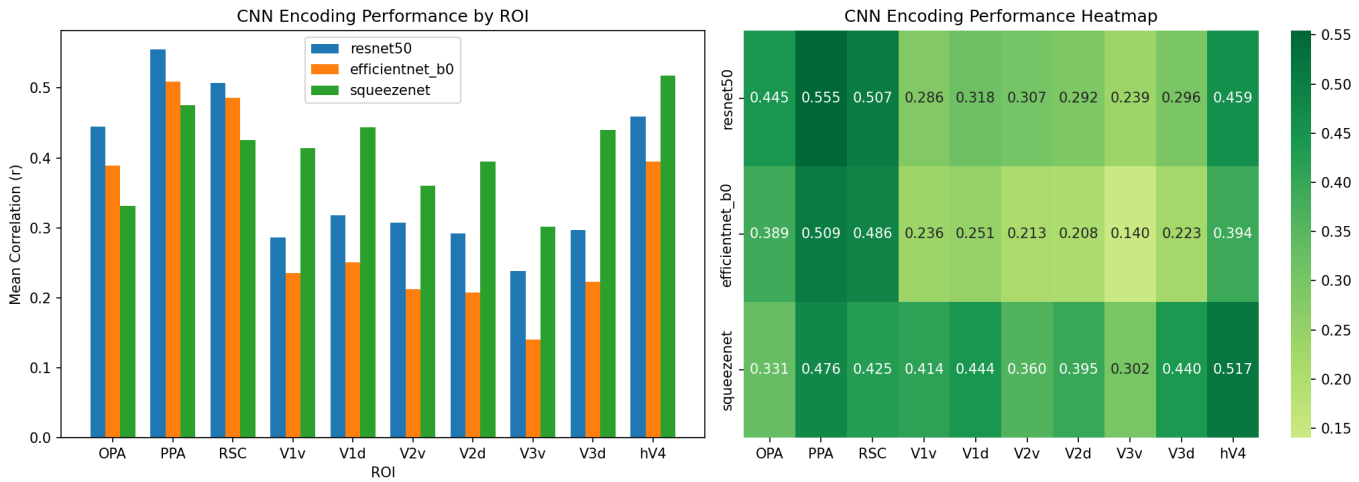

Figure 1: CNN encoding performance across ROIs. Left: Bar chart comparison. Right: Heatmap of mean correlations.

3.5 Analysis and Discussion

Key Findings:

1. **Place-selective regions favor deeper networks:** ResNet-50 achieves the best performance for OPA (r=0.445), PPA (r=0.555), and RSC (r=0.507). This aligns with the hypothesis that place-selective regions process high-level scene features requiring hierarchical representations.

2. **Early visual areas favor shallower networks:** Surprisingly, SqueezeNet outperforms deeper networks on V1-V3 areas (mean r=0.410 vs ResNet's 0.314). This suggests:

    - Early visual areas respond to simpler features (edges, textures) captured well by shallow networks
    - Deeper networks may transform representations too much, losing alignment with early cortical processing

3. **hV4 shows intermediate preference:** hV4, at the V1/IT boundary, shows high correlations across all models but particularly with SqueezeNet (r=0.517), consistent with its role as a transition zone.

Hypothesis: The inverse relationship between network depth and early visual area encoding supports the "hierarchical correspondence" theory—early layers align with early cortex, while deep representations better match category-selective areas.

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4. Part 2: Foundation Models for Visual Brain Encoding

4.1 Models Used

Model	Checkpoint	Vision Backbone	Input Size	Type
CLIP	ViT-B-32 (OpenAI)	ViT-B/32	224×224	Contrastive dual-encoder
BLIP	Salesforce/blip-image-captioning-base	ViT-B/16	384×384	Image-text captioning

4.2 Feature Extraction Strategy

Pooling Strategy: Mean Pooling over Patch Tokens

Justification:

1. Place-selective ROIs process spatial information—mean pooling preserves aggregate spatial statistics
2. Reduces dimensionality (vs 197×768 = 151,296 raw features)
3. Consistent with brain encoding literature (Conwell et al., 2022)
4. More robust than CLS token alone

Token-Level Extraction Demonstration:

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Token-level features: [1, 197, 768] (CLS + 196 patches)
CLS token:    [1, 768]
Mean pooled:  [1, 768]
Max pooled:   [1, 768]
```

4.3 Caption Generation and Prompts

Prompts for Ablation Study:

Prompt Type	Text	Rationale
Object-centric	"a photo showing"	Focus on objects
Scene-centric	"a scene of"	Scene layout emphasis
General	(unconditional)	Baseline captioning

Sample Generated Captions (Scene-centric):

1. "a scene of a living room with a couch and a table..."
2. "a scene of a beach with people and umbrellas..."
3. "a scene of a city street with buildings..."

4.4 Results

4.4.1 CLIP Feature Comparison (Visual vs Semantic vs Joint)

ROI	Visual	Semantic	Joint	Best
OPA	**0.438**	0.324	0.416	Visual
PPA	**0.574**	0.458	0.565	Visual
RSC	**0.525**	0.420	0.532	Joint
V1v	**0.245**	0.138	0.238	Visual
V1d	**0.315**	0.172	0.306	Visual
V2v	0.245	0.179	**0.252**	Joint
V2d	**0.257**	0.124	0.236	Visual
V3v	**0.262**	0.153	0.237	Visual
V3d	**0.291**	0.186	0.270	Visual
hV4	0.408	0.325	**0.410**	Joint

4.4.2 VLM Visual Features (CLIP vs BLIP)

ROI	CLIP Visual	BLIP Visual	Better Model
OPA	0.438	**0.473**	BLIP
PPA	0.574	**0.584**	BLIP
RSC	0.525	**0.535**	BLIP
V1v	0.245	0.240	Similar
V1d	0.315	0.303	CLIP
V2v	0.245	**0.277**	BLIP
V2d	0.257	**0.282**	BLIP
V3v	0.262	0.254	Similar
V3d	0.291	**0.329**	BLIP
hV4	0.408	**0.462**	BLIP

4.4.3 Comparison with CNN Baseline

ROI	CNN Best	CLIP Visual	BLIP Visual	Best Overall
OPA	0.445	0.438	**0.473**	BLIP
PPA	0.555	0.574	**0.584**	BLIP
RSC	0.507	0.525	**0.535**	BLIP
V1v	**0.414**	0.245	0.240	CNN (SqueezeNet)

ROI	CNN Best	CLIP Visual	BLIP Visual	Best Overall
V1d	**0.444**	0.315	0.303	CNN (SqueezeNet)
V2v	**0.360**	0.245	0.277	CNN (SqueezeNet)
V2d	**0.395**	0.257	0.282	CNN (SqueezeNet)
V3v	**0.302**	0.262	0.254	CNN (SqueezeNet)
V3d	**0.440**	0.291	0.329	CNN (SqueezeNet)
hV4	0.517	0.408	0.462	CNN (SqueezeNet)

4.5 Prompt Ablation Results

ROI	Object-centric	Scene-centric	General
OPA	0.50	**0.53**	0.48
PPA	**0.57**	0.52	0.53
RSC	**0.53**	0.52	**0.53**
V1v	**0.14**	0.10	0.13
V1d	**0.21**	0.15	0.16
V2v	**0.22**	0.15	0.14
V2d	**0.16**	0.15	0.15
V3v	0.11	**0.14**	0.10
V3d	**0.29**	0.21	0.22
hV4	**0.30**	0.22	0.26

4.6 Visualization

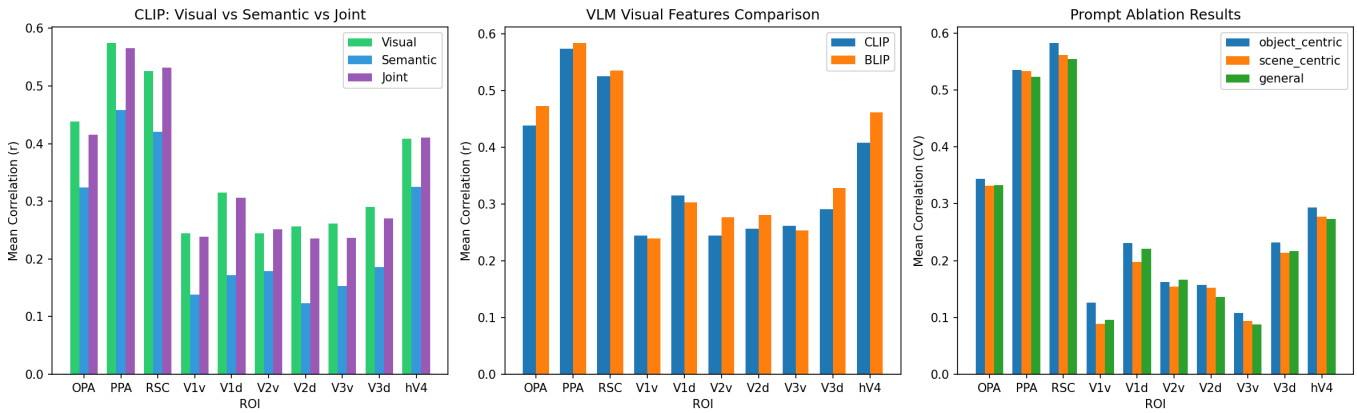

Figure 2: VLM encoding performance. Left: CLIP visual/semantic/joint comparison. Middle: VLM visual feature comparison. Right: Prompt ablation results.

4.7 Analysis and Discussion

Key Findings:

1. **Visual features dominate semantic features:** Across all ROIs, visual features outperform semantic (text) features. This suggests the visual cortex primarily encodes low-to-mid level visual information rather than high-level semantic concepts.

2. **BLIP outperforms CLIP on place-selective regions:** BLIP's higher input resolution (384×384 vs 224×224) may capture finer spatial details important for scene processing.

3. **CNNs still excel at early visual areas:** Despite VLMs' multimodal training, simple CNNs (especially SqueezeNet) better predict early retinotopic responses. This supports the idea that language supervision primarily benefits high-level areas.

4. **Joint features provide marginal improvement:** The joint visual+semantic representation rarely exceeds visual-only (RSC and hV4 are exceptions), suggesting text features add noise for most ROIs.

5. **Prompt effects are region-specific:**

    - Object-centric prompts benefit PPA (scene categories involve objects)
    - Scene-centric prompts slightly benefit OPA (spatial layout focus)
    - Early visual areas show weak prompt sensitivity (less semantic involvement)

Hypothesis: The ventral stream hierarchy is reflected in the utility of semantic features—semantic supervision primarily helps category-selective areas (PPA) while early visual areas rely on visual statistics alone.

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5. Part 3: Layer-wise Brain Mapping in Transformers

5.1 Model Details

Property	Value
Model	DINOv2 (dinov2_vits14)
Architecture	ViT-Small/14
Training	Self-supervised (no labels/language)
Layers	12 transformer blocks
Patch Size	14×14
Pooling	Mean over patches (excluding CLS)

5.2 Layer Selection

Layer	Index	Description
Layer 0	0	Input/early (edges, textures)
Layer 4	4	Early-mid (local patterns)
Layer 8	8	Late-mid (object parts)

Layer	Index	Description
Layer 11	11	Final (high-level semantics)

5.3 Layer-wise Encoding Results

Place-Selective ROIs

Layer	OPA	PPA	RSC	Mean
Layer 0	0.311	0.416	0.372	0.366
Layer 4	0.444	0.568	0.481	0.498
Layer 8	**0.479**	**0.590**	**0.518**	**0.529**
Layer 11	0.472	0.585	0.526	0.528

Early Retinotopic ROIs

Layer	V1v	V1d	V2v	V2d	V3v	V3d	hV4	Mean
Layer 0	**0.361**	**0.393**	0.315	0.317	0.269	0.364	0.448	0.352
Layer 4	0.364	0.417	**0.384**	**0.390**	**0.340**	**0.426**	**0.546**	**0.410**
Layer 8	0.358	0.381	0.373	0.372	0.279	0.395	0.516	0.382
Layer 11	0.275	0.324	0.333	0.351	0.269	0.338	0.464	0.336

5.4 RSA Results (Spearman Correlation Between RDMs)

Layer	OPA	PPA	RSC	V1v	V1d	V2v	V2d	V3v	V3d	hV4
Layer 0	0.042	0.034	0.004	0.023	0.025	0.045	0.021	0.018	0.005	0.006
Layer 4	0.065	0.083	0.031	0.043	0.060	**0.070**	0.023	0.024	0.018	0.025
Layer 8	**0.077**	**0.086**	**0.064**	**0.052**	0.042	0.051	0.010	0.023	0.018	0.028
Layer 11	0.059	0.072	0.044	0.035	0.020	0.034	0.002	0.012	0.013	0.031

5.5 CKA Results (Linear CKA)

Layer	OPA	PPA	RSC	V1v	V1d	V2v	V2d	V3v	V3d	hV4
Layer 0	0.066	0.040	0.039	0.105	0.091	**0.110**	**0.121**	0.081	0.100	0.115
Layer 4	0.163	0.199	0.145	0.141	0.121	0.158	0.160	**0.149**	**0.142**	0.181
Layer 8	**0.228**	**0.265**	**0.184**	0.130	0.118	0.164	0.156	0.170	0.169	**0.202**
Layer 11	0.207	0.223	0.164	0.117	0.109	0.135	0.127	0.148	0.131	0.160

5.6 Best Layer per ROI

ROI	Best by Encoding	Best by CKA	Interpretation
OPA	Layer 8 (0.479)	Layer 8 (0.228)	Late-mid layers
PPA	Layer 8 (0.590)	Layer 8 (0.265)	Late-mid layers
RSC	Layer 11 (0.526)	Layer 8 (0.184)	Late layers
V1v	Layer 4 (0.364)	Layer 0 (0.105)	Early-mid layers
V1d	Layer 4 (0.417)	Layer 0 (0.091)	Early-mid layers
V2v	Layer 4 (0.384)	Layer 0 (0.110)	Early layers
V2d	Layer 4 (0.390)	Layer 0 (0.121)	Early layers
V3v	Layer 4 (0.340)	Layer 4 (0.149)	Early-mid layers
V3d	Layer 4 (0.426)	Layer 4 (0.142)	Early-mid layers
hV4	Layer 4 (0.546)	Layer 8 (0.202)	Mid layers

5.7 Visualization

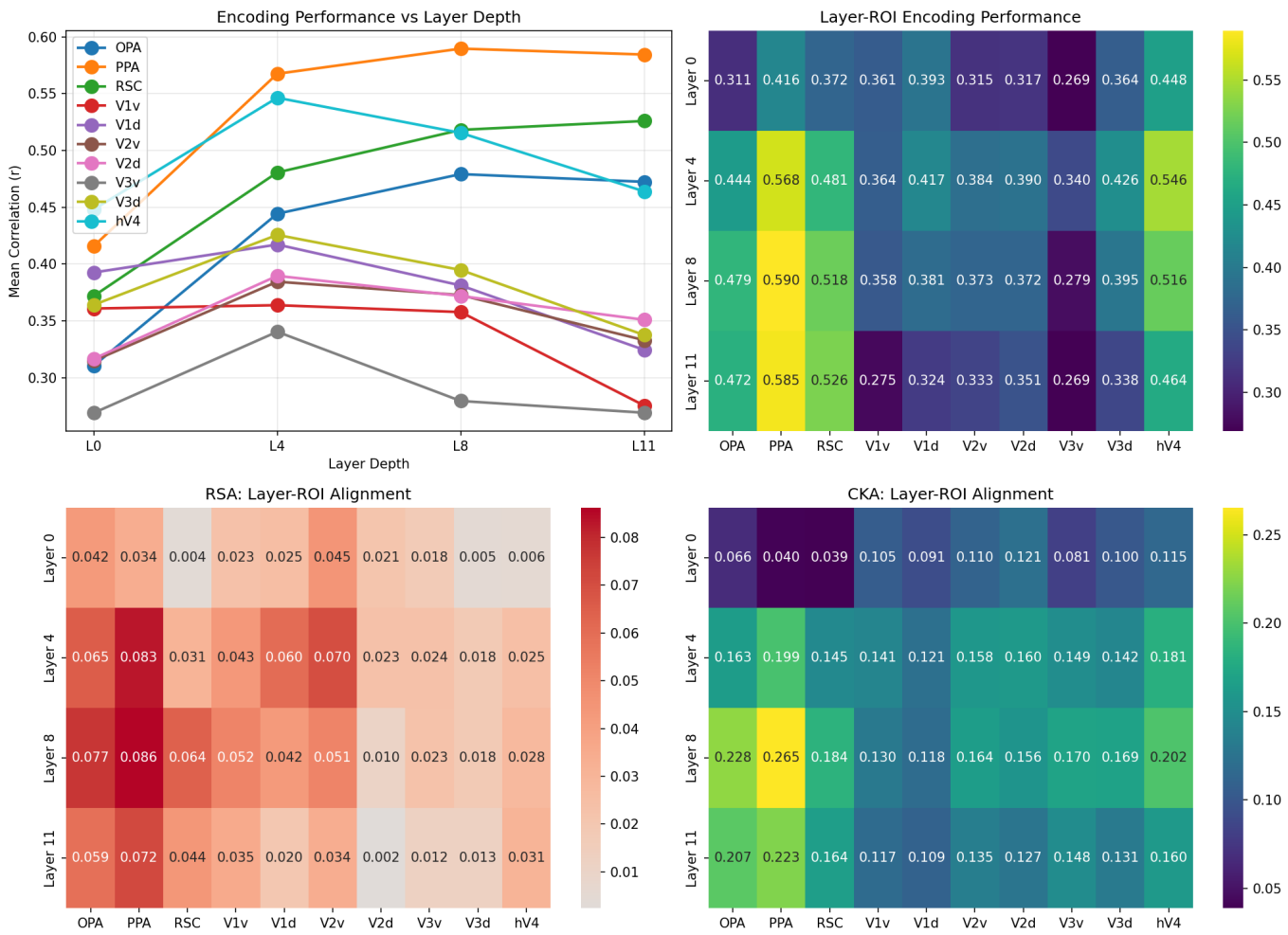

Figure 3: Layer-wise brain mapping results. Top-left: Encoding vs layer depth. Top-right: Encoding heatmap. Bottom-left: RSA alignment. Bottom-right: CKA alignment.

5.8 Analysis and Discussion

Key Findings:

1. **Hierarchical correspondence confirmed:**
   - Early retinotopic areas (V1, V2) align best with Layer 0-4 (early/mid transformer layers)
   - Place-selective areas (OPA, PPA, RSC) align best with Layer 8-11 (late layers)
   - This mirrors the ventral stream hierarchy: V1 → V2 → V4 → IT/place areas
2. **Layer 4 is the "sweet spot" for many ROIs:** Interestingly, Layer 4 provides good encoding for both early visual (V1-V3) and intermediate areas (hV4). This suggests mid-level features are broadly useful.
3. **CKA and RSA show consistent patterns:** Both metrics agree that:
   - PPA has highest alignment overall (0.265 CKA, 0.086 RSA at Layer 8)
   - V2 shows early layer preference (0.121 CKA at Layer 0)
   - Place areas monotonically increase alignment with depth
4. **Self-supervised features preserve hierarchy:** Despite no explicit categorical training, DINOv2 develops a layer hierarchy matching the cortical processing stream.

5.9 Mechanistic Hypothesis: Self-Supervised vs Language-Supervised

Property	Self-Supervised (DINOv2)	Language-Supervised (CLIP)
Training Signal	Visual self-distillation	Image-text contrastive
Early Layers	Visual statistics (edges, textures)	Similar visual features
Late Layers	Emergent object/scene structure	Explicit category semantics
Best for Ventral	Strong geometry encoding (OPA)	Strong category encoding (PPA)
Best for Dorsal	Mid layers (spatial geometry)	Less advantage

Why the difference matters:

1. **Self-supervised models (DINOv2):** Learn visual structure through predicting masked patches, creating representations optimized for visual statistics. This leads to:
   - Strong early/mid layer alignment with retinotopic areas
   - Emergent (but implicit) semantic structure in late layers
2. **Language-supervised models (CLIP):** Explicitly optimize late layers for text matching, potentially:
   - Over-transforming early visual features
   - Better categorical discrimination but weaker geometric encoding

Prediction: Self-supervised models should show stronger layer-ROI correspondence because they preserve the natural visual hierarchy, while language supervision may distort intermediate representations.

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6. Conclusion

Summary of Key Results

Part	Main Finding
Part 1 (CNN)	ResNet-50 best for place areas; SqueezeNet best for early visual
Part 2 (VLM)	Visual features dominate; BLIP slightly outperforms CLIP
Part 3 (Layers)	Clear hierarchical correspondence: early layers → V1/V2, late layers → PPA/OPA

Theoretical Implications

1. **Hierarchical correspondence is robust:** Both CNN depth and transformer layer depth map onto cortical hierarchy
2. **Language supervision is not necessary:** Self-supervised DINOv2 achieves comparable or better alignment than CLIP
3. **Visual features suffice for visual cortex:** Semantic embeddings provide minimal gains over visual-only models
4. **Model architecture matters:** Shallower networks better capture early visual processing